RUDRAKSH SYAL

Data Science Lead | Agentic AI & LLM Systems

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PROFESSIONAL PROFILE

Data science lead with 11+ years in Python, data science, and ML, building production ML, agentic AI, and LLM systems. Shipped an LLM tool-calling pricing layer on Anthropic Claude (AWS Bedrock) that moved from a parallel run to live use across 7 U.S. regions, with ~240 requests a day and 85–90% of recommendations accepted. Also built hybrid RAG over 4,000+ contracts and LLM extraction with human-in-the-loop review.

TECHNICAL SKILLS

Agentic AI & LLMs: LLM tool/function calling, Anthropic Claude (Sonnet, Haiku, Opus) via AWS Bedrock, OpenAI Agents SDK, LangChain/LangGraph, MCP, RAG, hybrid search, Llama 3.1 8B, structured extraction

Evaluation & Reliability: LLM-as-a-judge, golden sets, Precision@K/Recall@K, human-in-the-loop approval, guardrails, LangSmith tracing, parallel-run validation

Cloud, Data & MLOps: AWS (Bedrock Knowledge Bases, SageMaker, OpenSearch Serverless), Snowflake, Databricks, Azure, MLflow, Airflow, Python, SQL, R, Spark

ML & Statistics: XGBoost, Random Forest, forecasting, optimization, causal inference (BSTS, synthetic control), A/B testing, NLP

PROFESSIONAL EXPERIENCE

Data Science Lead | Hana E Solutions **Oct 2026–Present**

Remote, San Jose, CA

- **Agentic AI workflows:** Founding data science contributor building agentic AI workflows from scratch for two clients, mainly in pricing. Both are at the proof-of-concept stage.

Senior Data Scientist | Vista **Oct 2021–Jul 2026**

San Jose, CA (Remote)

Agentic AI, RAG & LLM Systems

- **Agentic pricing layer:** Built a custom Python LLM tool-calling layer (no framework) on Claude Sonnet, Haiku, and Opus via AWS Bedrock, on top of the Dynamic Pricing Engine. The LLM calls forecasting and Snowflake tools, explains pricing drivers from the structured results, and a pricing manager approves or rejects each recommendation.
- **Parallel run to adoption:** Ran a six-quarter program for 7 U.S. regions (5–8 fleet pricing managers and 2 revenue leaders each). A Q6 parallel run won buy-in, and the layer went live in Q7 at ~240 requests a day across 3 fleets. Rejections fell from ~30% to 10–15%.
- **Evaluation & guardrails:** Paired LLM-as-a-judge evaluations with approve/reject outcomes, using acceptance rate as the success metric. Wrote guardrails in custom Python and traced runs with LangSmith.
- **Hybrid RAG over contracts:** Built hybrid retrieval over 4,000+ contracts with Bedrock Knowledge Bases, OpenSearch Serverless, and a SageMaker-hosted Llama 3.1 8B. Evaluated it on golden question–clause sets (Precision@K/Recall@K).
- **LLM extraction to production:** Structured refund terms for ~600 operators into cancellation-policy badges with product and engineering. 84.7% were badge-ready automatically, and uncertain cases went to manual review.

Production ML & Pricing Systems

- **Dynamic Pricing Engine:** Led two data scientists to replace static pricing bands with a supply-demand equilibrium engine across seven U.S. regions, using six supply submodels and a bounded Brentq yield solver. One 43-aircraft fleet saw 28% fewer off-fleet bookings in month one.
- **EU cost model:** Built a gradient-boosted cost model, validated in shadow mode before relaunch. Margin went from −8% to +17%, and loss-making bookings fell from 55% to 20%.
- **Forecasting & propensity:** Raised booking forecast accuracy for 14 fleet categories from 71% to 78%. Built XGBoost lead-propensity scores integrated into Salesforce.

Senior Data Scientist | 84.51° (Kroger) **Jul 2019–Oct 2021**

Cincinnati, OH

- **Causal measurement:** Built BSTS and synthetic-control measurement for Quaker endcaps (4% lift vs. a 10% benchmark), which informed the decision to pause an Eastern-division rollout.
- **Space optimization:** Optimized bakery footage with Random Forest and integer programming. Nine pilot stores recorded 14% higher early sales (directional).

Data Scientist | Practo Technologies **Jun 2015–Apr 2018**

Bangalore, India

- **Analytics apps & NLP:** Built R/Shiny and Flask reporting apps that replaced a $4K/month-per-person vendor. Used sentiment NLP and A/B tests to find conversion drop-offs.

PROJECTS

- **Dynamic Pricing Engine** (personal rebuild on anonymised data)
  github.com/rudrakshsyal/dynamic_pricing_engine
  An OpenAI Agents SDK layer writes tool-bound revenue briefs, checked for citations and numbers. A budget-capped GPT-5.5 research agent was benchmarked against production-style ML forecasts; results were mixed, so the ML forecasts were kept.
- **Sidekick Agent** (personal project)
  github.com/rudrakshsyal/the_sidekick_agent
  A LangChain/LangGraph agent with MCP tools (Playwright browser, sandboxed filesystem), an evaluator loop with up to 3 retries, PII and model-call-limit guardrails, and human-in-the-loop approval.

LEADERSHIP, PATENT & EDUCATION

Leadership: Led two data scientists on pricing systems; started cross-functional learning sessions for 60+ colleagues.

Patent: U.S. Patent No. 12,314,757 B1 (2025): Efficient resource scheduling using adaptive scheduling criteria (co-inventor).

Purdue University — M.S., Business Analytics & Data Science (2019)

BITS Pilani — B.E. (Hons.) and M.Sc., Applied Mathematics (2015)